Building a reliable and reproducible scientific workflow using Nextflow



Nexflow Pipelines in HPC

Daniel Valle Millares
da.valle@ciberisciii.es
(Bioinformatics Platform - CIBERINFEC)

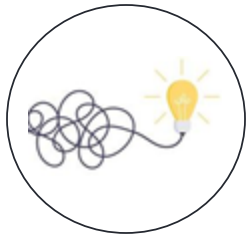
BU-ISCIII
octubre de 2026
2ª edición

Index

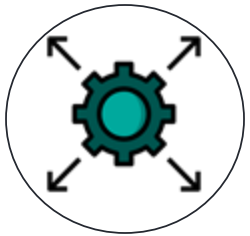
1. Scripting on the cluster — Slurm: sbatch & job arrays
2. Parallelization on our cluster — OpenMP vs MPI (when to use each)
3. **From scripts to workflows — building a reproducible pipeline (Nextflow preview)**
4. Wrap-up & Q&A

From scripts to workflows — building a reproducible pipeline (Nextflow preview)

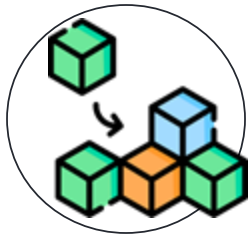
nextflow



Domain Specific Language



Scalability for Massive Data Analysis



Automation & Modularity



Reproducibility Across Platforms

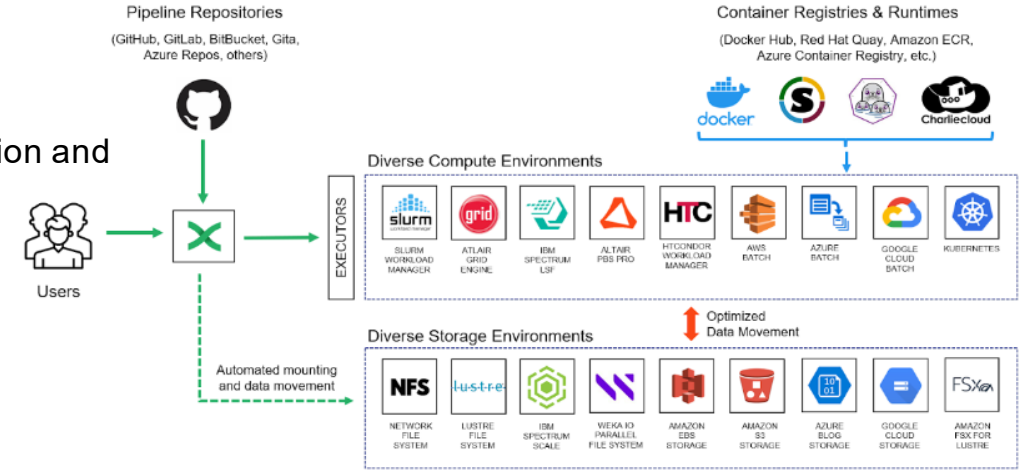
From scripts to workflows — building a reproducible pipeline (Nextflow preview)

Why Nextflow?

- Scripts → OK for one step.
- Workflows → connect many steps

- Nextflow adds a thin abstraction layer:
  - **You**: describe the pipeline and dataflow
  - **Nextflow**: handles SLURM job configuration and submission

- Benefits:
  - Not need to worry about:
    - Deploy slurm jobs
    - Parallelism
    - Resume tasks
  - Portability: minimum configuration to use your workflow in a laptop, HPC, Cloud…
  - Reproducibility (containers, safe resume, versioning…)



From scripts to workflows — building a reproducible pipeline (Nextflow preview)

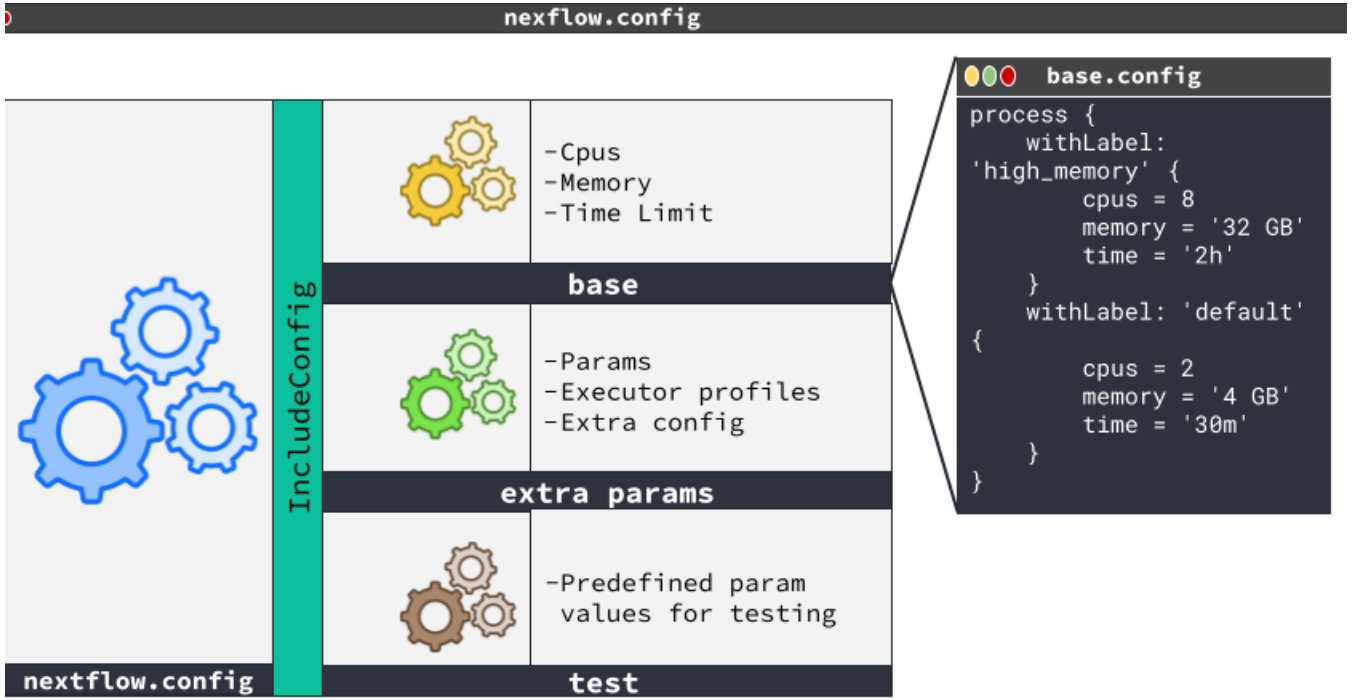
Nextflow on Slurm

1) Create/Use a nextflow pipeline



https://nf-co.re/demo/dev/

Nextflow Configuration Files



Nextflow Configuration Files

Nextflow on Slurm

1) Create/Use a nextflow pipeline
2) Create a **nextflow.config** file

```
process {
  executor = 'slurm'                                  // enviar tareas a Slurm
  queue    = 'middle_idx'                             // partición/cola por defect
  jobName  = { "$task.name-$task.hash" }              // nombre de job legible y
único
}
```

From scripts to workflows — building a reproducible pipeline (Nextflow preview)

Nextflow on Slurm

1) Create/Use a nextflow pipeline
2) Create a nextflow.config file
3) **Create sbatch script**

```
#!/bin/bash
#SBATCH --job-name=nf_demo
#SBATCH --partition=middle_idx
#SBATCH --time=12:00:00
#SBATCH --cpus-per-task=2                # Recursos SOLO para el
controlador de Nextflow
#SBATCH --mem=2G
#SBATCH --output=logs/%x-%j.out
#SBATCH --error=logs/%x-%j.err

# Carga las dependencias para ejecutar Nextflow
module purge
module load Nextflow/24.04.2
module load singularity/3.7.1

mkdir -p 01-nextflow-demo-results
# Ejecuta nf-core/demo (workflow preparado)
# Importante: le indicamos que lea el archivo de configuración
nextflow run nf-core/demo \
  -profile test,singularity \
  -c nextflow.config \
  --outdir 01-nextflow-demo-results \
  -resume
```

From scripts to workflows — building a reproducible pipeline (Nextflow preview)

Nextflow on Slurm

Monitor execution -- stdOUT

```
N E X T F L O W  ~  version 24.02.0
Launching `main.nf` [elegant_burnell] DSL2 - revision: a1b2c3d4f5
executor >  slurm (6)
[3a/5f1c2b] process > DOWNLOAD_TEST_DATA          [100%] 1 of 1 ✓
[b9/904773] process > FASTQC (sample_1)           [100%] 1 of 1 ✓
[6e/12ab34] process > FASTQC (sample_2)           [100%] 1 of 1 ✓
[2d/aa77cc] process > ECHO_HELLO (1)              [100%] 4 of 4 ✓
[7f/33dd44] process > MULTIQC                      [100%] 1 of 1 ✓

Completed at: 2025-09-01 11:42:17
Duration    : 2m 12s
CPU hours   : 0.1
Succeeded   : 8
```

From scripts to workflows — building a reproducible pipeline (Nextflow preview)

Nextflow on Slurm

Monitor execution -- squeue -me

```
JOBID PARTITION  NAME                      USER    ST  TIME  NODES NODELIST(REASON)
     9101201 short_idx  nf_demo                   daniel   R  0:01      1 ideafix03
     9101202 short_idx  FASTQC - a1b2c3d4         daniel   R  0:00      1 ideafix05
     9101203 short_idx  FASTQC - e5f6a7b8         daniel   R  0:00      1 ideafix08
     9101204 short_idx  FASTQC - 9c0d1e2f         daniel  PD  0:00      1 (Resources)
     9101205 short_idx  MULTIQC - 55aa33ff        daniel  PD  0:00      1 (Dependency)
     9101206 short_idx  ECHO_HELLO - deadbeef     daniel  CG  0:02      1 ideafix02
     9101207 short_idx  CLEANUP - f00dbabe        daniel  PD  0:00      1 (Dependency)
     9101208 short_idx  SUMMARY - cafed00d        daniel  PD  0:00      1 (Dependency)
```

Nextflow pipeline outputs

work/ (DON'T DELETE THIS)

- Stores all data generated during the workflow execution, including intermediate files, temporary files, and final results.
- This serves as a cache folder, enabling Nextflow's algorithm to resume executions without repeating steps that were successfully completed.

.nextflow/

- Stores information regarding Nextflow's execution

.nextflow.log

- Main log of Nextflow. Enables monitoring of the execution (log file).

The work folder

"work" [cache]

```
$: tree work/ -L 2

work
├── 7e
│   └── 30bfd15fadd931b83d1fda41d3df12
├── a9
│   └── f4efcb2891b829aa4fddf2713d643a
├── b9
│   └── 904773448b3b7bc9c5abb3e7767cf4
└── bf
    └── 74065bed76ef8c651238e460fd45ec
```

```
$: nextflow run hello.nf

N E X T F L O W  ~  version 23.10.1
Launching `main.nf` [voluminous_ardinghelli] DSL2 -
revision: b0eee95d4e
executor >  local (4)
[b9/904773] process > sayHello (4) [100%] 4 of 4 ✓
Ciao world!

Hello world!

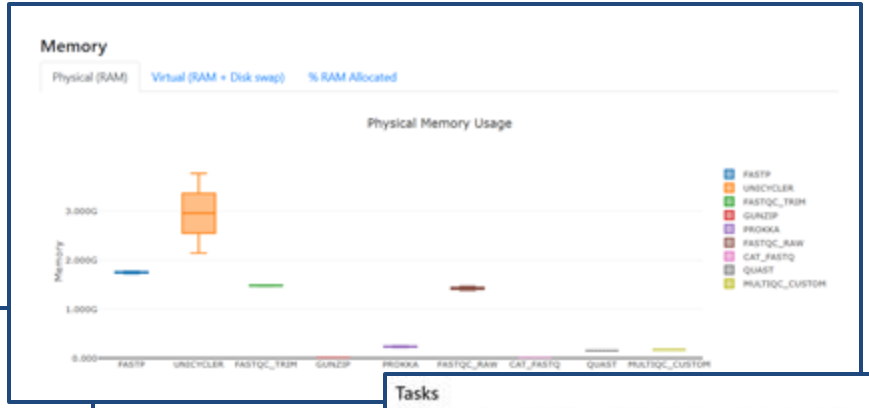
Bonjour world!

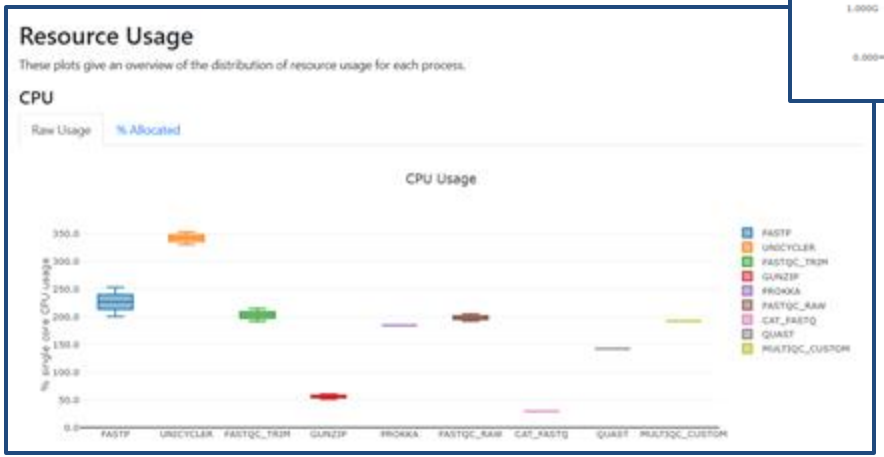
Hola world!
```

From scripts to workflows — building a reproducible pipeline (Nextflow preview)

Nextflow on Slurm

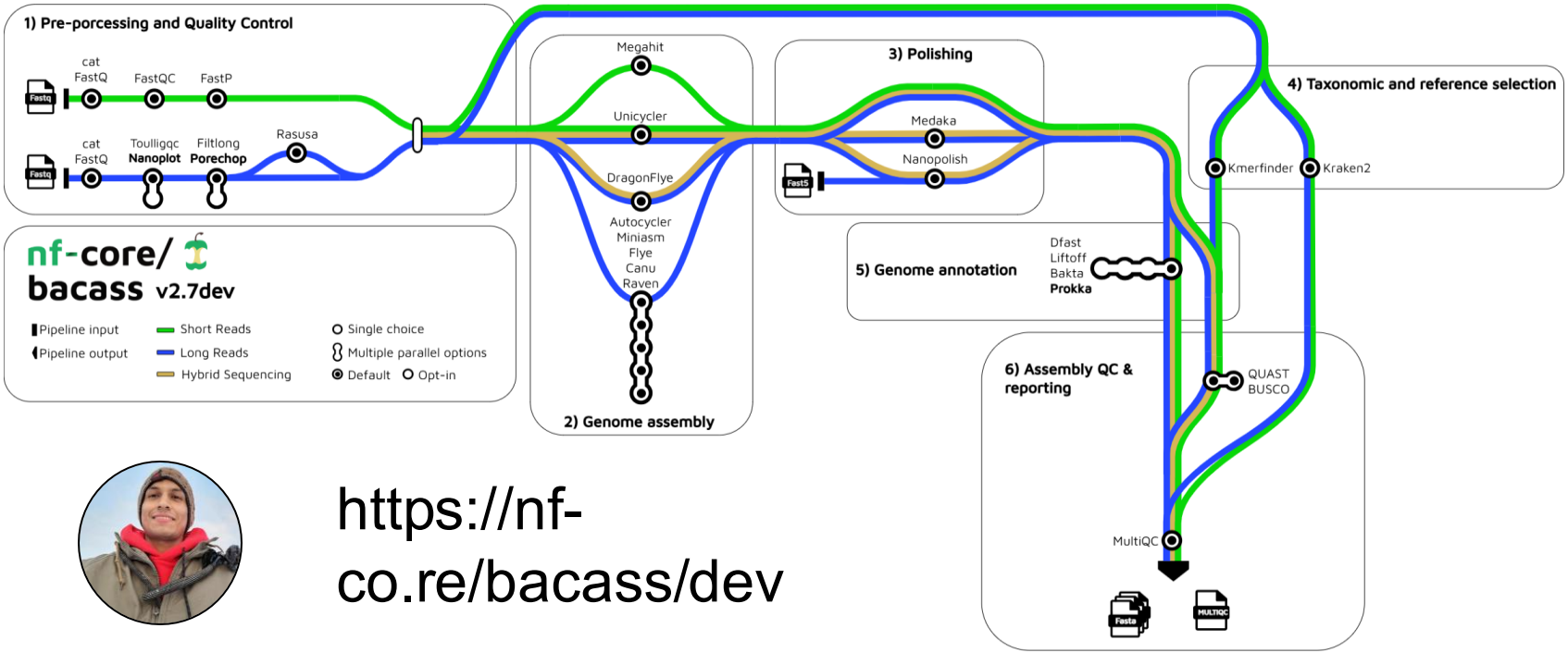
Monitor execution -- Fancy mode







Demo nf-core Bacass



https://nf-co.re/bacass/dev

Thank you for your attention

Questions?



HANDS-ON